OMNIKON

NATIONAL HACKATHON 2026

TEAM NAME

AlphaX duo

MEMBERS

Mounika A , Shanmugapriya S B

THEME

BioTech & HealthTech

PROBLEM STATEMENT

Conventional prosthetics are costly and often lack adaptive grip control, limiting daily functionality for underprivileged amputees. Although affordable and open-source designs exist, they are scattered and hard to discover. Amputees also struggle to find reliable local makers, get proper fitting guidance, and receive aftercare support. As a result, many functional low-cost limbs are either never accessed or abandoned after a short period.

PROBLEM STATEMENT

THE PROBLEM

Choosing a suitable prosthetic is challenging because:

- Users often don’t know which prosthetic design suits their needs
- Taking and understanding measurements is difficult without guidance
- Different designs have different mechanical characteristics
- Objective comparison of designs is hard
- Finding trustworthy and verified prosthetic makers is difficult
- Even after selection, simple fitting guidance is limited

Core Problem

- Users need a trusted and understandable pathway from assessment → selection → fitting guidance → reliable makers.

How It Addresses The Problem

- Solves discovery problem → users find suitable low-cost designs easily
- Solves fitting problem → clear guided process reduces pain and abandonment
- Solves trust problem → verified makers + quality signals
- Solves aftercare problem → continuous support and repair guidance

Innovation And Uniqueness

- Mechanical Quality Evaluation of every design (strength, grip usefulness, durability, suitability for Indian activities)
- Trust-first system designed specifically for low-cost market fears
- Fitting and training treated as a core feature, not an afterthought
- Strong emphasis on reducing prosthetic abandonment
- Built with deep understanding of both software and mechanical realities

Introducing ProstheticTrust:

- A digital platform that supports users throughout the prosthetic selection journey.

What we provide:

- Structured Assessment – Collect measurements and user requirements
- Design Recommendation – Match assessment data with design specifications
- Mechanical Quality Score – Clear scores for stability, strength, durability, adjustability & weight
- Fitting Guidance – Simple visual/video-based guidance
- Verified Makers – Discover trusted prosthetic makers
- Feedback & Rating – Build trust through real user experiences

Why ProstheticTrust?

Not just a prosthetic catalogue — it connects the full user journey:

Assess → Understand → Compare → Recommend → Guide → Connect → Feedback

What makes it different?

- Measurement-based decision support
- Mechanical transparency
- Built-in fitting education
- Trust system (Verified makers + Feedback + Ratings)

Features Implemented

Current Working Prototype

Authentication

- Firebase Email/Password Authentication
- Registration, Login, Logout
- User Profile
- Name, Phone, Role, Language
- Stored in Firestore

Assessment

- Affected side, Residual limb length, Forearm & Wrist circumference
- Hand length & width
- Input validation + Scrollable interface

Design Discovery

- Multiple design options
- Mechanical characteristics
- Design selection flow

Navigation Screens

Welcome → Need Selection → Limb Type → Assessment → Review → Recommendation → Guidance

Technical Architecture

Architecture Overview

Flutter App (Dart)
Authentication | Assessment | Review | Recommendations | Fitting Guidance | Maker Discovery | Feedback

Application Logic

- Recommendation Engine | Compatibility Evaluation | Mechanical Scoring | Input Validation
- Firebase Backend
- Firebase Authentication | Cloud Firestore | Secure user-specific data

Technologies

- Flutter
- Dart
- Firebase Authentication
- Cloud Firestore

PROGRESS MODE

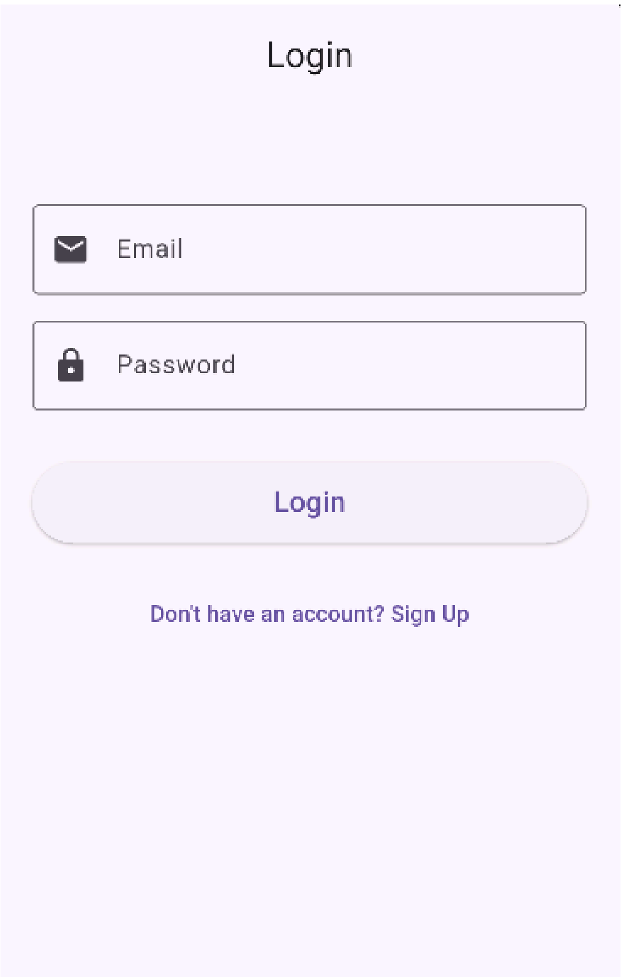
Development Progress

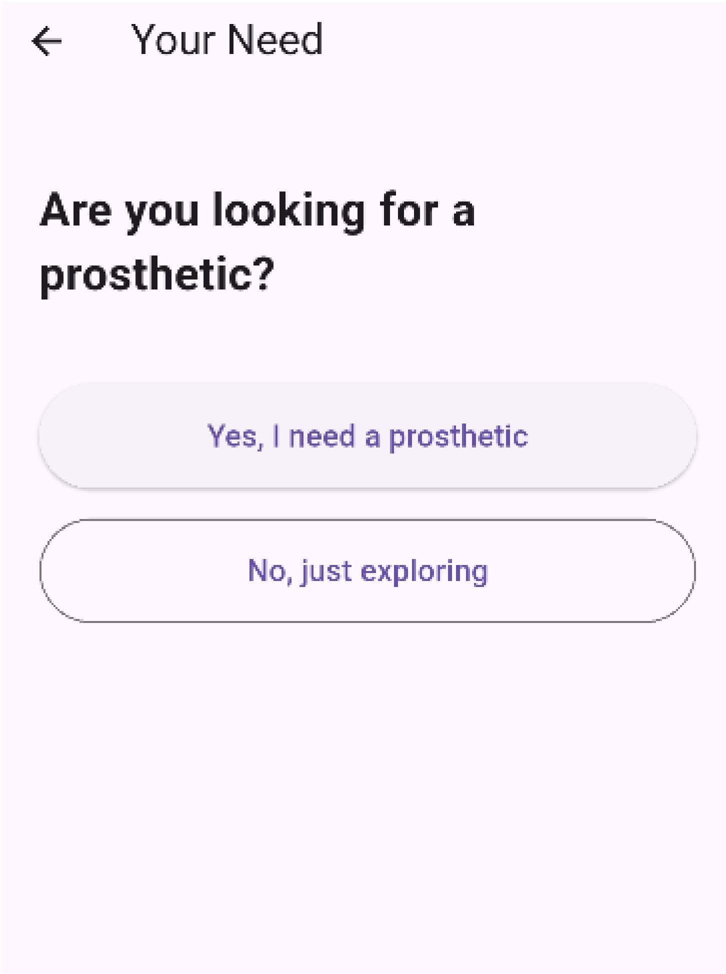
Completed

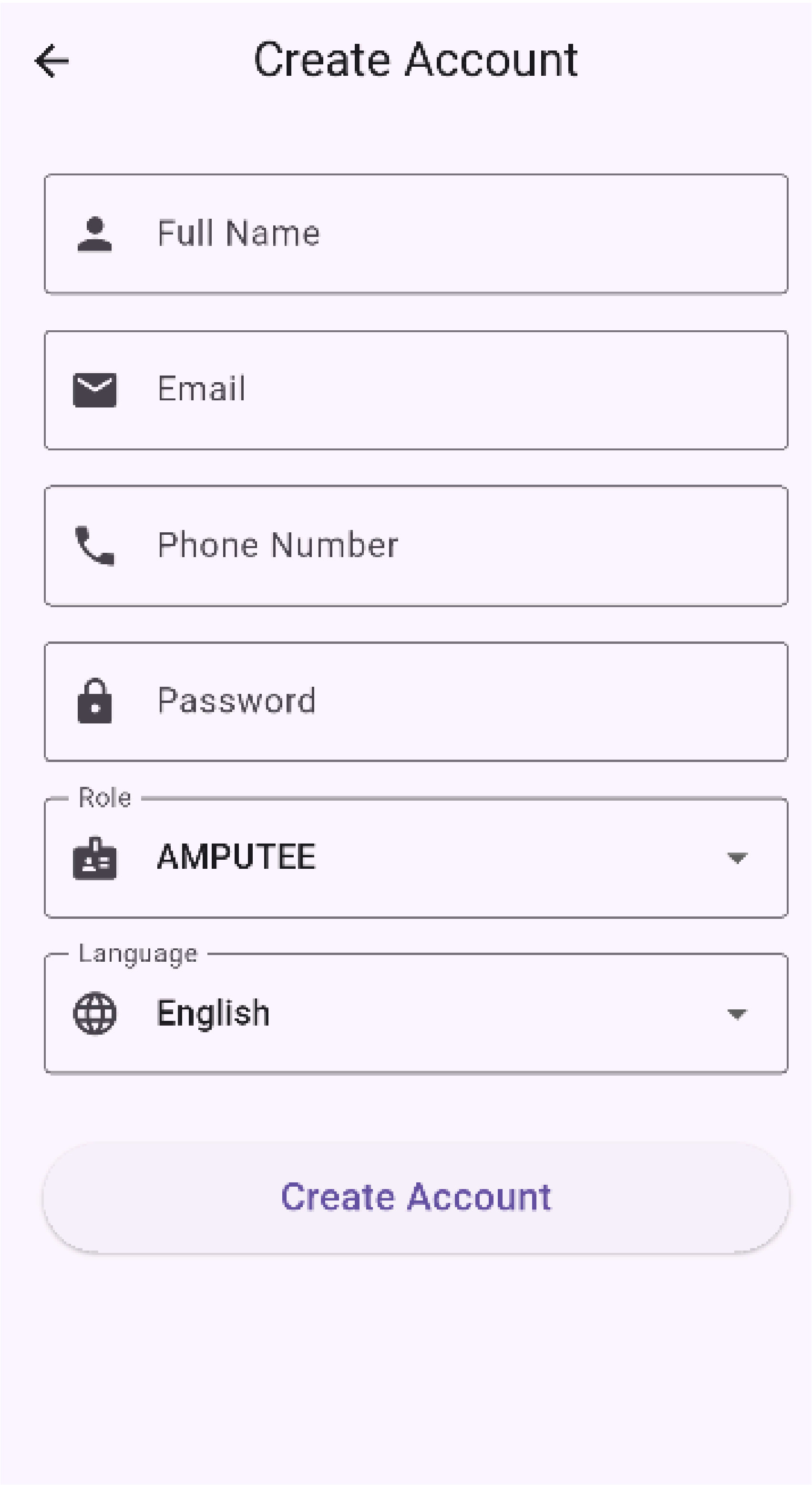
- Flutter project setup
- Firebase integration & Authentication
- User profile + Firestore storage
- Welcome, Need Selection, Limb Type flow
- Assessment screen with validation
- Review Assessment
- Design Recommendation flow
- Fitting Guidance structure

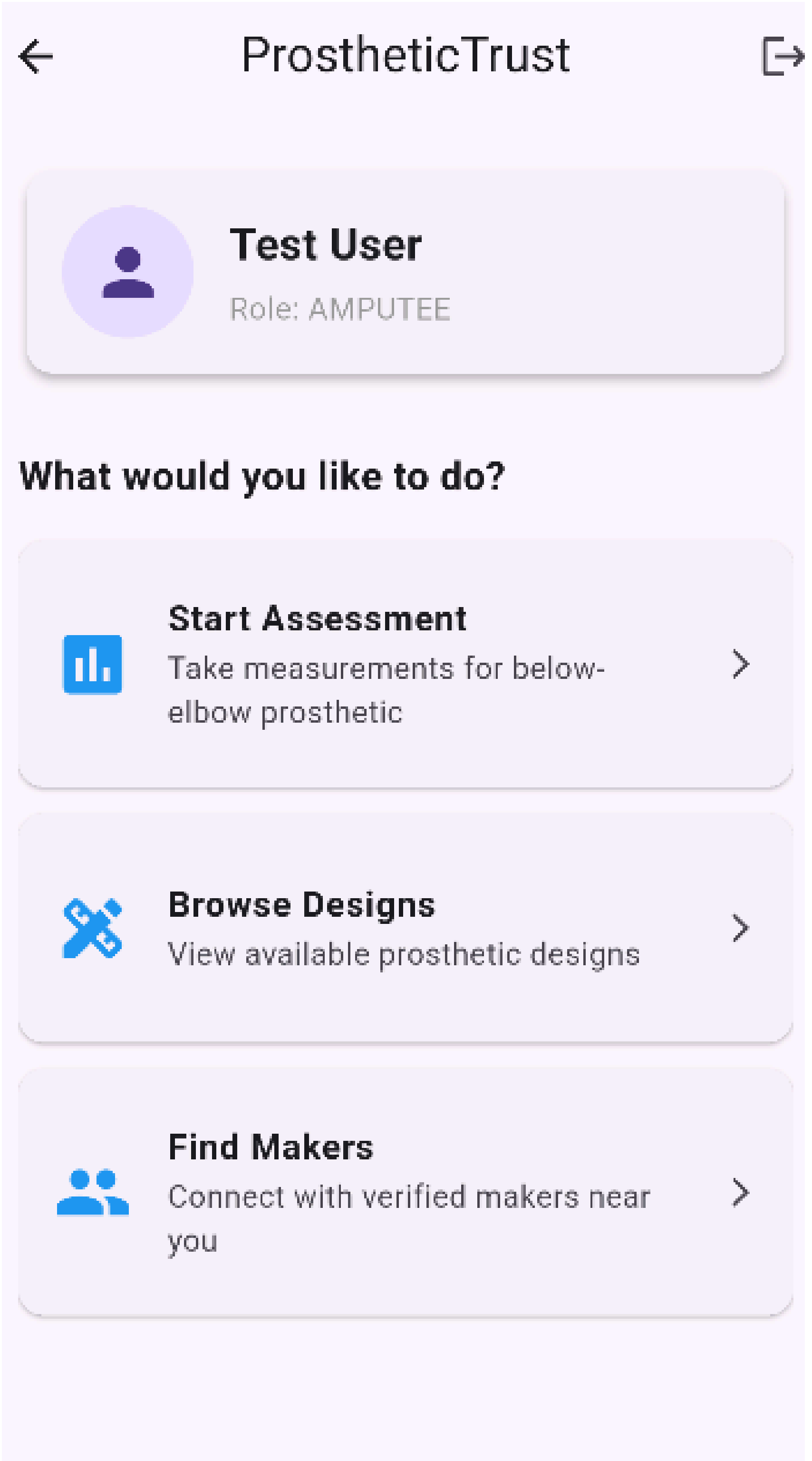
In Development

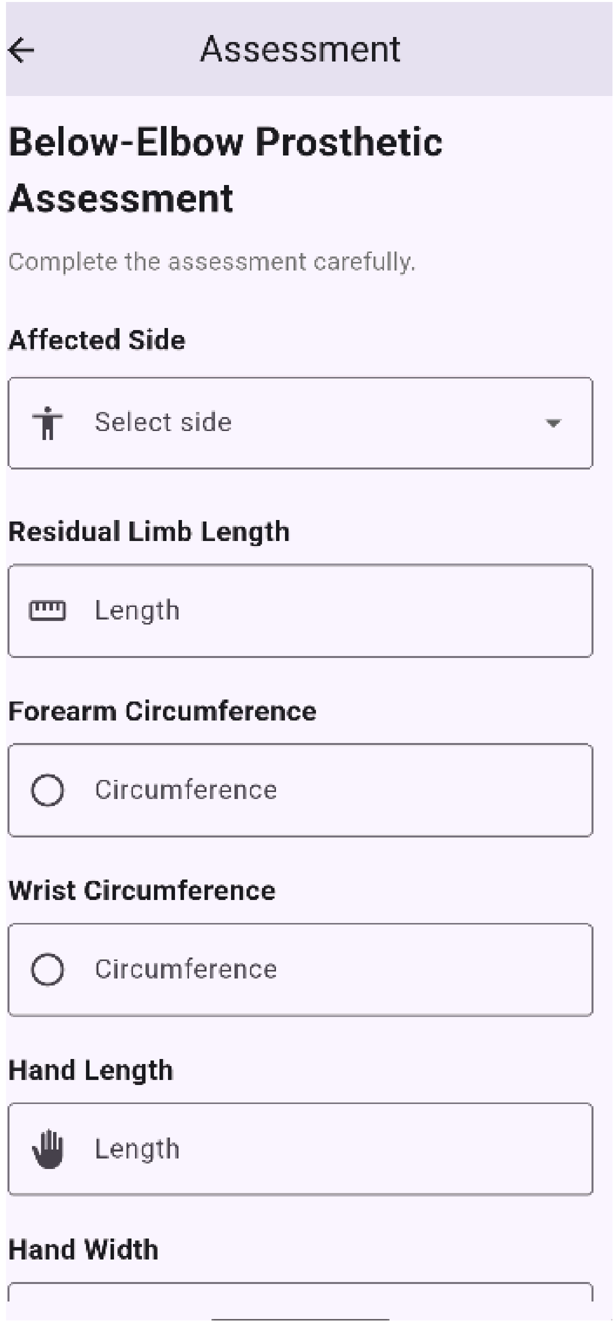
- Validated recommendation rules
- Mechanical Quality Score
- Measurement guidance videos
- Verified maker system
- Feedback & rating system


Login
Email
Password
Login
Don't have an account? Sign Up


Your Need
Are you looking for a prosthetic?
Yes, I need a prosthetic
No, just exploring


Create Account
Full Name
Email
Phone Number
Password
Role
AMPUTEE
Language
English
Create Account


ProstheticTrust
Test User
Role: AMPUTEE
What would you like to do?
Start Assessment
Take measurements for below-elbow prosthetic
Browse Designs
View available prosthetic designs
Find Makers
Connect with verified makers near you


Assessment
Below-Elbow Prosthetic Assessment
Complete the assessment carefully.
Affected Side
Select side
Residual Limb Length
Length
Forearm Circumference
Circumference
Wrist Circumference
Circumference
Hand Length
Length
Hand Width

HOW TO MEASURE THE BODY FOR FITTING A PROSTHETIC ARM (UPPER LIMB)

Accurate measurement ensures comfort, proper fit and better function.

GENERAL GUIDELINES

- Use a soft measuring tape.
- Measurements should be taken in centimetres (cm).
- The person should be relaxed and in a natural position.
- Measure the limb (affected side).

1 SHOULDER WIDTH

Measure the distance across the shoulders from one edge to the other.

2 CHEST GIRTH

Measure the circumference of the chest, just under the armpits.

3 UPPER ARM LENGTH

Measure from the shoulder point (acromion) to the end of the residual limb.

4 UPPER ARM CIRCUMFERENCE

Measure the circumference of the upper arm at the widest part.

5 ELBOW WIDTH

Measure the width across the elbow (widest part).

6 ELBOW TO END LENGTH

Measure from the elbow joint to the end of the residual limb.

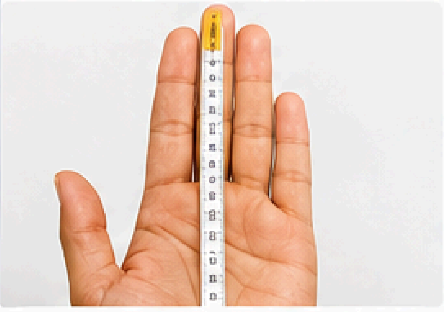
7 FOREARM CIRCUMFERENCE

Measure the circumference of the forearm (widest part).

8 WRIST CIRCUMFERENCE (If Present)

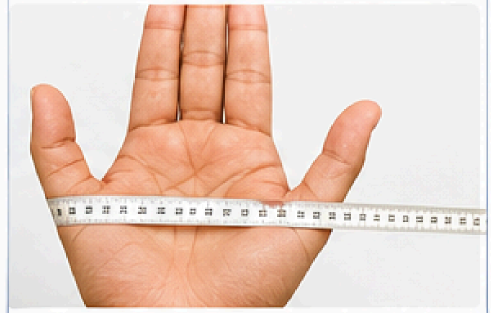
Measure the circumference of the wrist.

9 HAND LENGTH (If Present)



Measure from the wrist crease to the tip of the middle finger.

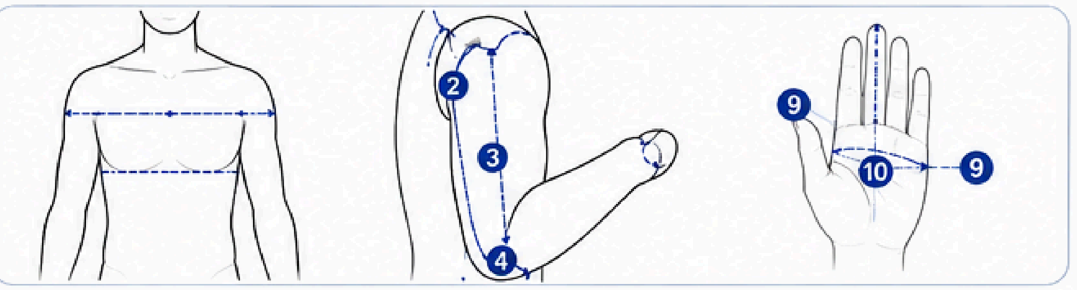
10 HAND WIDTH (If Present)



Measure the width of the hand across the knuckles.

EXTRA INFORMATION TO COLLECT

- Age
- Weight
- Height
- Cause of limb loss
- Time since amputation
- Activity level & daily needs



NOTE

Take all measurements carefully. A small measurement error can affect the fit and comfort of the prosthetic limb.

STEP 1 — SELECT YOUR AMPUTATION LEVEL

■ Above the elbow	■ Below the elbow
■ At the elbow	■ At the wrist

STEP 2 — BODY & LIMB MEASUREMENTS

1. Shoulder Width Measure the distance across the shoulders from one prominent shoulder point to the other.	Enter: ____________ **cm**
2. Chest Girth Measure the circumference of the chest, just under the armpits.	Enter: ____________ **cm**
3. Upper Arm Length Measure from the shoulder point (acromion) to the end of the residual limb.	Enter: ____________ **cm**
4. Upper Arm Circumference Measure the circumference of the upper arm at the widest part.	Enter: ____________ **cm**
5. Elbow Width Measure the width across the elbow at the widest part.	Enter: ____________ **cm**
6. Elbow-to-End Length Measure from the elbow joint to the end of the residual limb.	Enter: ____________ **cm**
7. Forearm Circumference Measure the circumference of the forearm at the widest part.	Enter: ____________ **cm**
8. Wrist Circumference If the wrist is present, measure the circumference of the wrist.	Enter: ____________ **cm**
9. Hand Length If the hand is present, measure from the wrist crease to the tip of the middle finger.	Enter: ____________ **cm**

STEP 3 — PHOTOGRAPH THE RESIDUAL LIMB

Upload two clear photographs so the prosthetic specialist can understand the shape of the limb.

FRONT VIEW	SIDE VIEW
Show the entire residual limb from the front. Photo: ____________________	Show the entire residual limb from the side. Photo: ____________________

STEP 4 — ADDITIONAL INFORMATION

Age	__________ years	**Height**	__________ cm
Weight	__________ kg	**Affected side**	■ Left ■ Right ■ Both
Time since amputation	______________	**Activity level**	■ Low ■ Moderate ■ High
Cause of amputation	____________________		

CURRENT NEEDS — Select all that apply

■ Daily activities	■ Work
■ School / education	■ Sports / physical activities
■ Cosmetic appearance	■ Other: ______________

PAIN OR SENSITIVE AREAS

____

____

STEP 5 — REVIEW BEFORE SUBMITTING

✓	All measurements are entered in cm.
✓	The tape was not pulled tightly.
✓	Measurements were taken in a relaxed position.
✓	Circumferences and widths were measured at the correct locations.
✓	Front and side photographs are clear and show the residual limb.

SAVE & CONTINUE

CHALLENGES FACED

Challenges & Our Approach

- Measurement Accuracy

Challenge: Users may enter incorrect measurements
Approach: Validation + Instructions + Review screen + Professional verification note

- Recommendation Reliability

Challenge: Arbitrary rules can give poor recommendations
Approach: Validated design specifications + Clear compatibility rules

- Trust

Challenge: Users need confidence in recommendations and makers
Approach: Verified makers + Transparent scoring + Feedback & ratings

- Accessibility

Challenge: Users have different technical abilities
Approach: Simple UI + Visual guidance + Multilingual support + Planned voice assistance

FUTURE ROADMAP

Roadmap

Phase 1 – Functional Prototype (Completed)
Core user journey • Authentication • Assessment • Navigation
Phase 2 – Smart Recommendation (In Progress)
Validated design specifications • Compatibility engine • Explainable mechanical score
Phase 3 – Guidance
Measurement videos • Fitting guidance • Multilingual support • Voice assistance
Phase 4 – Trust Network
Verified makers • Maker profiles • Ratings • User feedback
Phase 5 – Scale
Offline support • Regional maker discovery • Improved accessibility • Data-driven improvements

EXPECTED IMPACT

For Users

- Easier understanding of prosthetic options
- Structured assessment
- Better design comparison
- Accessible educational guidance
- Easier discovery of trusted makers

For Makers

- Structured user requirements
- Better communication
- Opportunity to build trust through verification and feedback

Long-Term Vision

- Build a trusted digital ecosystem connecting users, prosthetic designs, guidance, makers and feedback.